

# REPORT

## Food Production Visualization Using Power BI

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☒ Project Title: Food Production Visualization

### Introduction

This project focuses on visualizing food production data from Kaggle using Power BI, enabling users to explore trends in a simple and interactive way. By using different chart types, the project presents meaningful insights from raw data without requiring any programming or preprocessing. The goal is to make data interpretation easier for stakeholders such as researchers, policymakers, and students.

### Project Overview

The project involves directly importing food production data from Kaggle into Power BI and creating interactive visualizations. It helps users quickly understand production patterns by crop, country, and time period.

### Objective

Visualize Kaggle food production data using Power BI  
Use multiple chart types to represent data meaningfully  
Enable easy filtering and analysis through slicers and visuals

### Data Source

Source: Kaggle  
Format: CSV  
Tool Used: Power BI Desktop  
Dataset Content: Food production by country, crop, and year

### Visualization Components

**Column Chart** Used to compare total production quantity for different countries or crops

**Card** Displays key summary values such as total production, number of countries, or crop count

**Ribbon Chart** Shows the change in rank or position of top crops or countries over time

**Donut Chart** Represents the proportion of each crop in overall production visually

**Gauge** Highlights performance toward a production goal or threshold (e.g., % of expected output)

**Area Chart** Displays production trends over time with a focus on volume and growth

**Funnel Chart** Represents stages such as global to regional to crop-level breakdown (or filtering impact)

## Dashboard Features

Interactive slicers for year, country, and crop

Dynamic visuals that change based on filters

Clear comparison tools like column and ribbon charts

Summary cards for quick insights

Visual storytelling through a combination of modern chart types

## Key Insights

Countries like India and China dominate in rice and wheat production

Donut and ribbon charts highlight major crops and their shifting ranks

Funnel view narrows down from global to country-specific insights

Gauge chart shows how much of the goal is achieved in production levels

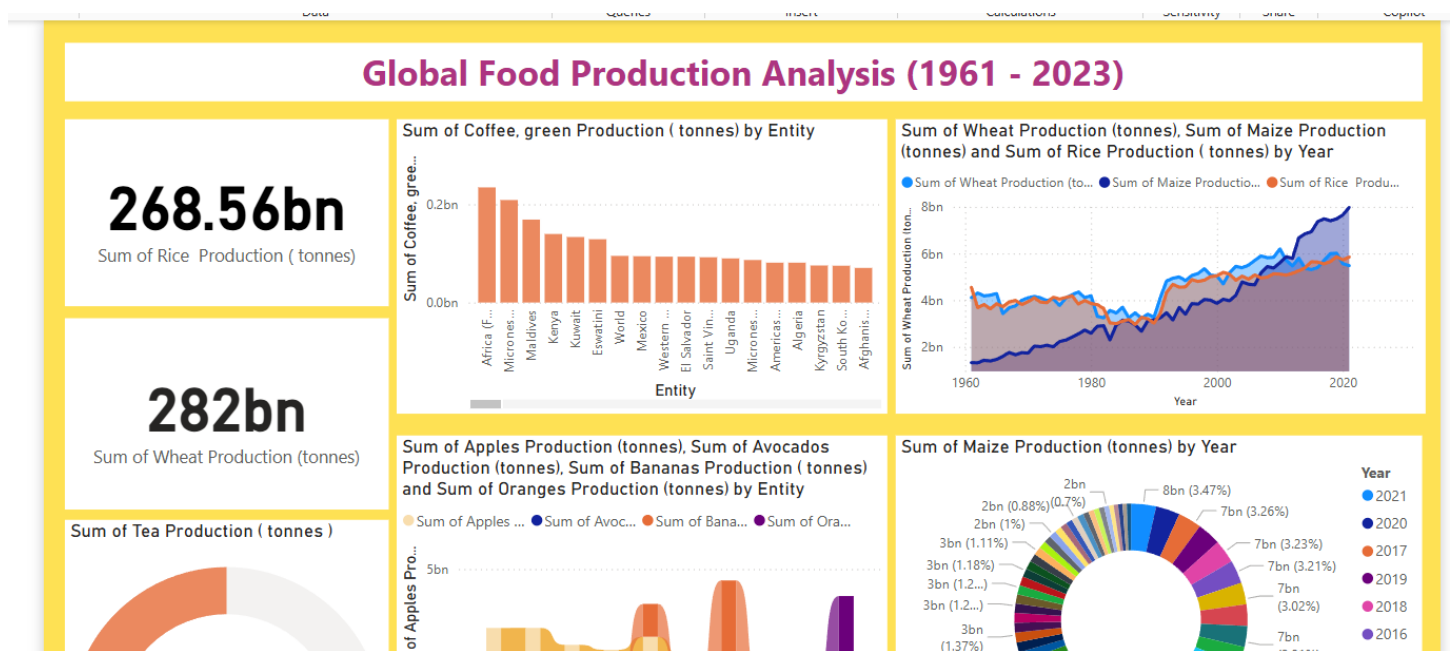
## Conclusion

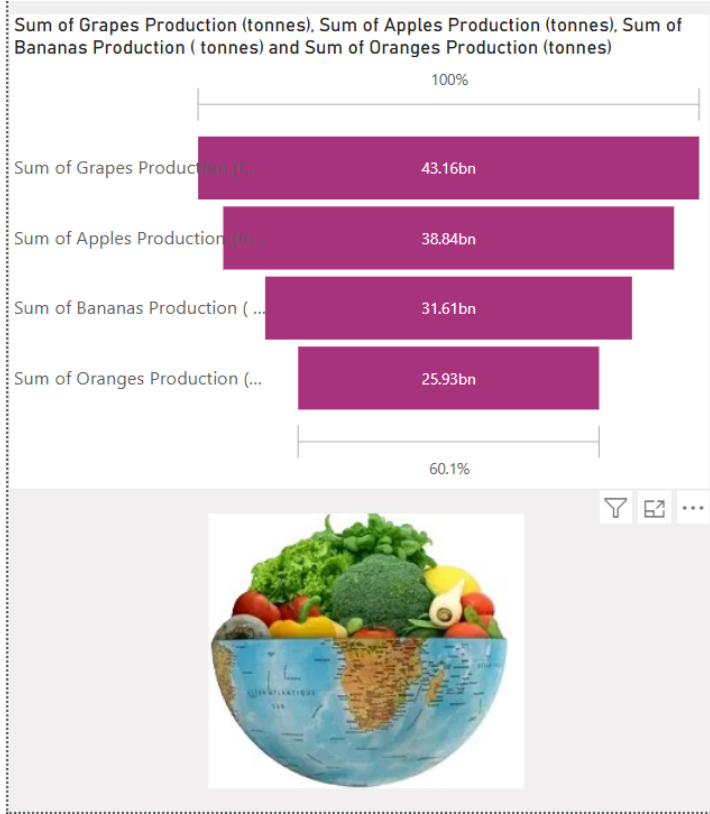
This Power BI project effectively turns raw Kaggle data into actionable insights using a wide variety of visuals. With just drag-and-drop tools and basic filtering, it empowers users to explore food production patterns interactively, supporting quick decision-making and deep analysis—without writing a single line of code.

## Attachments

Kaggle\_Dataset.csv (Raw dataset)

## Dashboard Screenshots(Power Bi Desktop)





**REPORT**

The total rice production globally from 1961 to 2023 is 269 billion tonnes.

The total wheat production globally from 1961 to 2023 is 282 billion tonnes.

The total tea production globally from 1961 to 2023 is 2 billion tonnes.

Africa, America, and Asia lead in the production of green coffee, with Africa being the top producer followed by America.

Wheat, maize, and rice production have all shown a steady increase from 1961 to 2023, with wheat production showing the most significant rise over the years.

Apples, avocados, bananas, and oranges are produced in varying quantities by different entities, with countries like Europe and Asia showing significant production volumes.

Maize production has consistently increased over the years, with notable jumps around the late 1980s and continuing into the 2000s.

Grapes have the highest total production at 43 billion tonnes, followed by apples (39 billion tonnes), bananas (32 billion tonnes), and oranges (26 billion tonnes).

**Kaggle\_Dataset.csv (Raw dataset)**

| Clipboard                 |            | Font |        | Alignment |          | Number  |         | Styles |         | Cells |       |        |       |        |       |
|---------------------------|------------|------|--------|-----------|----------|---------|---------|--------|---------|-------|-------|--------|-------|--------|-------|
| C1                        |            |      |        |           |          |         |         |        |         |       |       |        |       |        |       |
| Maize Production (tonnes) |            |      |        |           |          |         |         |        |         |       |       |        |       |        |       |
|                           | A          | B    | C      | D         | E        | F       | G       | H      | I       | J     | K     | L      | M     | N      | O     |
| 6                         | Afghanista | 1965 | 720000 | 380000    | 10067913 | 2282000 | 2559608 | 72418  | 3169104 | 13200 | 51000 | 73067  | 13900 | 132000 | 13700 |
| 7                         | Afghanista | 1966 | 720000 | 337000    | 10863614 | 2033000 | 2690984 | 90272  | 3214807 | 14000 | 51000 | 77180  | 8411  | 136000 | 15300 |
| 8                         | Afghanista | 1967 | 768000 | 396000    | 12123091 | 2280000 | 2580187 | 86111  | 3556573 | 14000 | 57000 | 80884  | 14301 | 147000 | 18500 |
| 9                         | Afghanista | 1968 | 773000 | 402000    | 12840044 | 2354000 | 2799236 | 100736 | 3601209 | 14400 | 57000 | 90459  | 9100  | 150000 | 18700 |
| 10                        | Afghanista | 1969 | 785000 | 407000    | 14496418 | 2454000 | 3001560 | 114363 | 3760912 | 14800 | 60000 | 89000  | 8100  | 154000 | 18900 |
| 11                        | Afghanista | 1970 | 667000 | 366000    | 16413323 | 2081000 | 3111482 | 121398 | 4558892 | 14000 | 55000 | 97211  | 15160 | 144000 | 18300 |
| 12                        | Afghanista | 1971 | 670000 | 350000    | 14232062 | 1915000 | 3180161 | 119437 | 4349264 | 11200 | 50000 | 97545  | 7120  | 167000 | 14400 |
| 13                        | Afghanista | 1972 | 720000 | 400000    | 11274815 | 2450000 | 3335950 | 151578 | 4130662 | 13600 | 35000 | 103891 | 12000 | 158000 | 17800 |
| 14                        | Afghanista | 1973 | 760000 | 420000    | 11600265 | 2700000 | 3359676 | 157667 | 4274725 | 15200 | 58000 | 106221 | 15320 | 165000 | 18500 |
| 15                        | Afghanista | 1974 | 770000 | 420000    | 12099241 | 2750000 | 3693404 | 153865 | 4981184 | 16000 | 51600 | 135960 | 8850  | 182000 | 18700 |
| 16                        | Afghanista | 1975 | 780000 | 435000    | 13487679 | 2850000 | 4441689 | 154792 | 5406268 | 16000 | 60000 | 151128 | 7980  | 194500 | 18800 |
| 17                        | Afghanista | 1976 | 800000 | 448000    | 11424861 | 2936000 | 4329267 | 160838 | 5725577 | 16000 | 68100 | 177483 | 9330  | 354000 | 17600 |
| 18                        | Afghanista | 1977 | 760000 | 400000    | 11006217 | 2652000 | 4353471 | 198731 | 5331775 | 14400 | 64000 | 214047 | 5900  | 200000 | 10400 |
| 19                        | Afghanista | 1978 | 780000 | 428000    | 10671106 | 2813000 | 4433050 | 205356 | 5738778 | 14000 | 64000 | 308253 | 10870 | 250000 | 15200 |
| 20                        | Afghanista | 1979 | 760000 | 439000    | 10360042 | 2663000 | 4639119 | 206032 | 5338973 | 16800 | 64000 | 327037 | 11800 | 257000 | 14500 |
| 21                        | Afghanista | 1980 | 752000 | 415000    | 10773930 | 2550000 | 4821352 | 200307 | 5234967 | 17200 | 70000 | 339734 | 9900  | 266000 | 15000 |

